

PMS Dashboard

Project Status & Release Readiness

An evidence-based technical program audit of software completion, security posture, database readiness, and release forecasting for management review.

 **CURRENT PROJECT STATUS: AT RISK (RELEASE BLOCKED BY SEC-01 & REPORT BUILDER DATA PARITY)**

REPORT DATE
July 23, 2026

AUDIT VERSION
v2.5.0-AUDIT

PREPARED BY
Senior Technical Program Manager

TARGET RELEASE
Controlled Pilot (Aug 2026)

1. Executive Summary

Overall Status, Primary Assessment & Management Decision Summary

OVERALL STATUS

AT RISK

Blocked by 2 critical items

PRODUCT COMPLETION

88%

13 Teams Configured

AUTOMATED TEST PASS RATE

100%

156 Vitest / 347 Pytest Clean

TARGET PILOT WINDOW

Aug 15–25

Controlled Pilot (3 Teams)

CRITICAL MANAGEMENT TAKEAWAY & RELEASE BLOCKERS

The PMS Dashboard software is functionally rich, beautifully styled, and technically mature. All automated unit test suites pass with 100% success (156 Vitest tests, 347 Pytest backend tests, 0 ESLint errors, 0 TypeScript errors). However, **immediate production release is BLOCKED** by two mandatory pre-requisites:

1. SEC-01 Critical Security Blocker: Historical Git commits contain un-rotated sensitive credentials and test data. A mandatory repository history rewrite (BFG/Gitleaks) and secret rotation must be executed prior to public exposure or production deployment.

2. Report Builder Data Parity Failure: A formal audit ('REPORT_BUILDER_PHASE_1_CLOSURE.md') declared Phase 1 incomplete due to data model discrepancies between Web UI, Insights, and generated PDF reports (e.g. zero-target null conversions).

Major Accomplishments & Capabilities Verified

- Unified KPI Scoring Engine:** Verified math handling uncapped raw achievement vs weight-capped contribution points across all operational teams.
- Config-Driven Onboarding:** Successfully deployed JSON-driven team onboarding supporting 13 operational teams across Egypt & UAE.
- Security Hardening (SEC-02 to SEC-08):** Sanitized current HEAD, implemented explicit CORS allowlists, Redis password auth, and Nginx 25MB upload protection.
- 16:9 Story Builder PDF Engine:** Implemented vector PDF export with SHA-256 integrity hashes and 6 seeded system story templates.
- Level-Scoped Access Control:** Deployed granular assignment API ('UserTeamAssignment') matching user authorization by team and performance level.

Project Context & Architecture Summary

COMPONENT	TECHNOLOGY STACK & ARCHITECTURE
Frontend	React 19, TypeScript ~6.0, Vite 8, Tailwind CSS v4, Zustand, Recharts, Framer Motion.
Backend	Python 3.11, FastAPI, SQLAlchemy 2.0 ORM, Python-SocketIO, Pydantic v2.
Database	PostgreSQL 15 (v3.0 DDL), Range Partitioning (2020–2030), RLS Scoping Policies.
Infrastructure	Docker Compose (Prod), Nginx SPA/WS Proxy, Redis 7 (Auth), Prometheus/Grafana.

2. Phase-by-Phase Status & Delivery Matrix

Detailed Audit Classification across 16 Technical & Functional Workstreams

Workstream Phase	Status	Comp %	Completed Scope	Remaining Scope	Verification Evidence
1. Foundation & Architecture	COMPLETED	100%	Monorepo scaffolding, FastAPI + React 19 shell, setup scripts, env templates.	None.	SYSTEM_AUDIT_REPORT.md
2. Database & Migrations	PARTIALLY VERIFIED	85%	PostgreSQL 15 schema v3.0, 31 Alembic migrations, RLS policies, partitioning.	Staging PostgreSQL dry-run migration.	Alembic head c1a8f6d2e4b7
3. Authentication & RBAC	VERIFIED	90%	JWT tokens, RBAC roles (Admin, Manager, Executive), level-scoped team access.	Password reset policy on first login.	test_auth_hardening.py
4. Workbook Upload Intake	VERIFIED	85%	Excel parser, signature validation (OLE/ZIP), bounded reads, Nginx 25MB limit.	Async background queue for large files.	upload_security.py
5. KPI Builder & Config	VERIFIED	90%	JSON-driven team configs for 13 teams, versioning table, weight sum triggers.	GUI config editor for non-technical users.	NEW_TEAM_ONBOARDING.md
6. Employee Performance	COMPLETED	95%	Employee directory, historical profile, uncapped vs capped contribution scoring.	Minor zero-target visual edge cases.	KpiBreakdownPanel.test.tsx
7. Team Dashboard	COMPLETED	95%	Multi-period filters, roster view toggles, call-center & back-office views.	None.	TeamDashboardView.tsx
8. Management Overview	COMPLETED	90%	Managerial & Corporate overview, manager selector bar, position hierarchy.	Cross-departmental drill-downs.	ManagerSelectorBar.test.tsx
9. Balanced Scorecard	COMPLETED	90%	4 perspectives (Financial, Customer, Internal, Learning), BSC snapshot engine.	Multi-year BSC trend comparison.	management_kpi_config tables
10. Strategy Map	VERIFIED	85%	Strategic objective mapping, position-based title resolution.	Interactive target node editing.	SYSTEM_STATUS.md
11. Insights & Planning	VERIFIED	80%	Insight drawer, canonical narrative engine, planning workspace (plans, milestones).	Persisted root cause register.	PlanningView.test.tsx
12. Report Builder	AT RISK	65%	5-step story builder UI, 6 templates, dynamic page expansion, 16:9 PDF export.	Resolve Phase 1 Audit failure (zero-target nulls, evidence parity).	REPORT_BUILDER_PHASE_1_CLOSURE.md
13. Security Hardening	AT RISK	75%	SEC-02 to SEC-08 resolved (CORS allowlist, upload security, Redis auth, 0 vulnerabilities).	SEC-01: Git history rewrite & secret rotation.	SYSTEM_AUDIT_REPORT.md

WORKSTREAM PHASE	STATUS	COMP %	COMPLETED SCOPE	REMAINING SCOPE	VERIFICATION EVIDENCE
14. Containerization	VERIFIED	85%	Multi-stage Dockerfiles (non-root execution), docker-compose.prod.yml, health probes.	Multi-worker WS pub/sub scaling.	docker-compose.prod.yml
15. Pilot & UAT Prep	PENDING	40%	Pre-flight verification checklist, 13 team onboarding definitions, handover zip archives.	Staging deployment, UAT user provisioning.	production-checklist.md
16. Production Deployment	PENDING	20%	Deployment guides (VPS, Cloud PaaS, Compose), Systemd service templates, Nginx SSL.	Production VPS setup, SSL certbot, DNS.	DEPLOYMENT_GUIDE.md

3. Current Operational Position & Scope Breakdown

What Is Fully Functional vs What Scope Remains Incomplete

WHAT IS CURRENTLY USABLE & VERIFIED (DEMONSTRABLE TODAY)

- **Full Interactive Frontend Shell:** Authenticated route navigation, role-aware sidebar, dynamic theme colors, responsive tables.
- **13 Onboarded Operational Teams:** Inbound, Outbound, Inbound UAE, Pre-Approvals IP Offshore, Sales, Pharmacy, Coding, CSR, Submission, Re-Submission, Pre-Approvals OP Dubai, Pre-Approvals IP Final Dubai, Marketing.
- **Scoring & Roster Views:** Live toggle between employee roster grids, performance level filters, and month selection.
- **Balanced Scorecard & Strategy Map:** Full workspace displaying Financial, Customer, Process, and Growth perspectives tied to live database snapshots.
- **Real-Time Push Notifications:** Socket.IO integration delivering targeted notifications to Admin and Manager team rooms.
- **Report Story Builder UI:** 5-step interactive workflow, drag-and-drop narrative blocks, 16:9 PDF preview and vector generation.

REMAINING UNFINISHED SCOPE BEFORE PRODUCTION

- **Product / Data Scope:** Fix zero-target null handling in Report Builder, align ReportingEvidenceService calculation parity with Dashboard UI.
- **Technical / Security Scope:** Execute SEC-01 Git history rewrite using BFG/Gitleaks, rotate all secrets in historical `.env` files.
- **Infrastructure / Ops Scope:** Implement Celery/Redis Queue background processing for Excel uploads; setup automated daily PostgreSQL logical backups via cron.
- **Pilot / UAT Scope:** Deploy staging environment on VPS, conduct controlled UAT with 3 initial teams (Inbound, Outbound, Pre-Approvals).

4. Detailed Release Readiness Assessment

Multi-Dimension Evaluation of Security, Data, Infrastructure & Operational Readiness

1. SECURITY & COMPLIANCE

AT RISK (75%)

- HEAD code is secure (CORS allowlist, Redis auth, 0 vulnerabilities).
- **BLOCKER:** Git history contains historical secrets (SEC-01).

2. DATABASE & DATA MODEL

PARTIALLY VERIFIED (85%)

- Schema v3.0 DDL verified with partitioning & RLS.
- **ACTION:** Staging Postgres migration dry-run required.

3. APPLICATION ARCHITECTURE

VERIFIED (90%)

- FastAPI + React 19 architecture verified.
- 100% test suite pass rate (156 Vitest / 347 Pytest).

4. INFRASTRUCTURE & CONTAINERS

VERIFIED (85%)

- Multi-stage Dockerfiles enforce non-root execution.
- docker-compose.prod.yml verified with health probes.

5. OPERATIONS & BACKUPS

PENDING (40%)

- Backup scripts written (`backup-db.sh`).
- **ACTION:** Cron mapping & recovery audit pending VPS setup.

6. PILOT & USER ACCEPTANCE

PENDING (30%)

- Pre-flight checklist defined.
- **ACTION:** Controlled pilot with 3 initial teams pending.

5. Executive Risk Register & Mitigation Strategy

Prioritized Assessment of Data, Security, Infrastructure & Delivery Risks

ID	RISK & CATEGORY	PROB	IMPACT	CURRENT MITIGATION STATUS	REQUIRED MANAGEMENT ACTION	RELEASE EFFECT
BLK-01	SEC-01: Sensitive Credentials in Historical Git Commits Security & Compliance	HIGH	CRITICAL	HEAD branch sanitized. Private runtime JSON un-tracked.	Execute BFG/Gitleaks history rewrite; rotate all DB, JWT, and Redis credentials immediately.	BLOCKS RELEASE
BLK-02	Report Builder Phase 1 Calculation Discrepancies Product Data Parity	HIGH	HIGH	Documented in REPORT_BUILDER_PHASE_1_CLOSURE.md (FAIL closure).	Refactor Report Builder to consume canonical ReportingEvidenceService and preserve zero-target nulls.	BLOCKS REPORT UI
RSK-01	Synchronous Upload Ingestion Thread Blocking Performance & Scale	Medium	Medium	Bounded chunk reads and Nginx 25MB body limit enforced.	Implement Celery/Redis Queue background worker task for large Excel uploads.	Affects Scaling
RSK-02	Unverified Staging PostgreSQL Migration Graph Database Integrity	Medium	High	31 Alembic migrations verified on local Postgres PMS_Sys.	Perform full migration dry-run on staging PostgreSQL VPS prior to UAT.	UAT Prerequisite
RSK-03	Single-Worker WebSocket Concurrency Bottleneck Infrastructure	Low	Medium	Socket.IO scoped rooms active. In-memory adapter running.	Deploy Gunicorn/Uvicorn multi-worker process manager with Redis pub/sub adapter.	Post-Pilot Scale

RISK SUMMARY & GOVERNANCE OVERVIEW

Out of 5 identified program risks, **2 are critical blockers** that prevent immediate deployment. The security risk (BLK-01) carries zero tolerance for enterprise production. The Report Builder data parity risk (BLK-02) threatens executive data credibility if unaddressed. Both can be fully remediated within a **10-business-day focused sprint**.

6. First Release Forecast & Delivery Timeline

Evidence-Based Schedule for Controlled Pilot, UAT, and Full Production Rollout



Release Release Strategy & Windows

RELEASE DIMENSION	TARGET WINDOW / SPECIFICATION
Recommended First Release	CONTROLLED PILOT RELEASE
Pilot Scope	3 Initial Teams: Inbound, Outbound, Pre-Approvals IP Offshore
Technical Readiness Date	August 7, 2026
Pilot / UAT Readiness Date	August 21, 2026
Full Production Readiness	September 15, 2026
Most Likely Release Window	September 1 – September 15, 2026
Forecast Confidence Level	High (Conditional on SEC-01 & Report fixes)

Critical Path Execution Sequence

- Git History Rewrite & Secret Rotation (SEC-01) [3 Days]:** Run BFG Repo-Cleaner/Gitleaks sweep, rotate PostgreSQL/Redis/JWT secrets.
- Report Builder Refactoring [5 Days]:** Unify calculation logic around `ReportingEvidenceService` and preserve zero-target nulls.
- Staging Deployment & Migration Dry-Run [3 Days]:** Deploy Docker Compose on staging VPS, run Alembic PostgreSQL migrations.
- Controlled Pilot UAT Execution [10 Days]:** Conduct UAT with team leads from Inbound, Outbound, and Pre-Approvals.
- Production Promotion & Full Rollout [5 Days]:** Onboard remaining 10 teams, configure automated backups & SSL.

7. Recommended Next Actions & Required Management Decisions

Actionable Roadmap for Immediate Execution & Sign-off

Immediate Next Actions (Next 48 Hours & 7 Days)

- **Next 48 Hours:** Approve SEC-01 Git history rewrite maintenance window and authorize credential rotation.
- **Next 48 Hours:** Assign Backend Lead to implement `ReportingEvidenceService` refactoring for Report Builder.
- **Next 7 Days:** Execute BFG Repo-Cleaner sweep across all branch histories.
- **Next 7 Days:** Deploy staging environment on Ubuntu VPS with Docker Compose and run Postgres migration dry-run.

Pre-Pilot & Pre-Production Actions

- **Before Pilot / UAT:** Provision UAT accounts with level-scoped permissions (`performance\_level`).
- **Before Pilot / UAT:** Conduct onboarding workshop for Inbound, Outbound, and Pre-Approvals team leads.
- **Before Production:** Implement Celery/Redis Queue background worker queue for Excel file ingestion.
- **Before Production:** Map `scripts/backup-db.sh` to daily cron and configure offsite snapshot replication.

8. Management Decisions Required

Key Sign-offs Needed from Executive Leadership

REQUIRED MANAGEMENT DECISION	DECISION OWNER	REQUIRED SIGN-OFF DATE
1. Authorize SEC-01 Git History Rewrite & Secret Rotation	CTO / IT Security Lead	July 26, 2026
2. Approve Controlled Pilot Release Strategy (3 Initial Teams)	VP of Operations / Product Sponsor	July 28, 2026
3. Allocate Staging & Production VPS Hosting Budget (Hetzner / AWS)	IT Infrastructure Director	July 30, 2026

FINAL AUDIT CONCLUSION & RECOMMENDATION

Status: AT RISK (RELEASE BLOCKED BY SEC-01 & REPORT BUILDER DATA PARITY)

The PMS Dashboard project represents a major technical achievement. Its user experience, scoring engine, performance architecture, and automated test coverage are outstanding. By approving the 48-hour security maintenance window and authorizing the Report Builder data model refactoring, executive management can confidently target a **Controlled Pilot Release in August 2026** followed by **Full Production Rollout by September 15, 2026**.

Audit Sign-off: Senior Technical Program Manager & Delivery Auditor

Signature: _____

Date: July 23, 2026